

## SAFETY DATA SHEET

Creation Date 04-February-2010

Revision Date 20-September-2024

Revision Number 10

### 1. Identification

**Product Name** Iron(II) chloride tetrahydrate

**Cat No. :** AC205080000; AC205080010; AC205080050; AC205080051; AC205082500

**CAS-No** 13478-10-9  
**Synonyms** Ferrous chloride tetrahydrate

**Recommended Use** Laboratory chemicals.  
**Uses advised against** Food, drug, pesticide or biocidal product use.

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

Acros Organics  
One Reagent Lane  
Fair Lawn, NJ 07410

##### **Manufacturer**

Fisher Scientific Company  
One Reagent Lane  
Fair Lawn, NJ 07410  
Tel: (201) 796-7100

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

##### **WHMIS 2015 Classification**

Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                          |            |
|------------------------------------------|------------|
| <b>Corrosive to metals</b>               | Category 1 |
| <b>Acute oral toxicity</b>               | Category 4 |
| <b>Serious Eye Damage/Eye Irritation</b> | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

May be corrosive to metals

Harmful if swallowed

Causes serious eye damage



### Precautionary Statements

#### Prevention

Keep only in original container

Wash face, hands and any exposed skin thoroughly after handling

Do not eat, drink or smoke when using this product

Wear protective gloves/protective clothing/eye protection/face protection

#### Response

IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing

Immediately call a POISON CENTER/doctor

Rinse mouth

Absorb spillage to prevent material damage

#### Storage

Store in corrosive resistant polypropylene container with a resistant inliner

#### Disposal

Dispose of contents/container to an approved waste disposal plant

## 3. Composition/Information on Ingredients

| Component                                        | CAS-No     | Weight % |
|--------------------------------------------------|------------|----------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | 13478-10-9 | <=100    |
| Iron chloride (FeCl <sub>2</sub> )               | 7758-94-3  | -        |

## 4. First-aid measures

|                                        |                                                                                                                                                                                                                                                                                                                   |
|----------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>General Advice</b>                  | If symptoms persist, call a physician.                                                                                                                                                                                                                                                                            |
| <b>Eye Contact</b>                     | Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.                                                                                                                                                                                                   |
| <b>Skin Contact</b>                    | Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.                                                                                                                                                                                                 |
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur.                                                                                                                                                                                                      |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Get medical attention if symptoms occur.                                                                                                                                                                                                             |
| <b>Most important symptoms/effects</b> | None reasonably foreseeable. Causes severe eye damage. Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                                                                                                                                             |

## 5. Fire-fighting measures

**Suitable Extinguishing Media** CO<sub>2</sub>, dry chemical, dry sand, alcohol-resistant foam.

**Unsuitable Extinguishing Media** No information available

**Flash Point** No information available  
**Method -** No information available

**Autoignition Temperature** No information available

**Explosion Limits**

**Upper** No data available

**Lower** No data available

**Sensitivity to Mechanical Impact** No information available

**Sensitivity to Static Discharge** No information available

**Specific Hazards Arising from the Chemical**

The product causes burns of eyes, skin and mucous membranes.

**Hazardous Combustion Products**

Iron oxides. Hydrogen chloride.

**Protective Equipment and Precautions for Firefighters**

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

**NFPA**

**Health**  
3

**Flammability**  
0

**Instability**  
0

**Physical hazards**  
N/A

## 6. Accidental release measures

**Personal Precautions** Use personal protective equipment as required. Ensure adequate ventilation. Avoid dust formation.

**Environmental Precautions** Should not be released into the environment. Do not allow material to contaminate ground water system. Do not flush into surface water or sanitary sewer system.

**Methods for Containment and Clean Up** Sweep up and shovel into suitable containers for disposal. Keep in suitable, closed containers for disposal.

## 7. Handling and storage

**Handling** Wear personal protective equipment/face protection. Do not get in eyes, on skin, or on clothing. Ensure adequate ventilation. Avoid ingestion and inhalation. Avoid dust formation.

**Storage.** Keep containers tightly closed in a dry, cool and well-ventilated place. Incompatible Materials. Strong oxidizing agents.

## 8. Exposure controls / personal protection

**Exposure Guidelines**

| Component                                        | Alberta                  | British Columbia                                      | Ontario TWAEV            | Quebec                     | ACGIH TLV                | OSHA PEL                           | NIOSH                           |
|--------------------------------------------------|--------------------------|-------------------------------------------------------|--------------------------|----------------------------|--------------------------|------------------------------------|---------------------------------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1.0 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup> | REL = 1 mg/m <sup>3</sup> (TWA) |
| Iron chloride (FeCl <sub>2</sub> )               | TWA: 1 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup><br>STEL: 2 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | TWA: 1.0 mg/m <sup>3</sup> | TWA: 1 mg/m <sup>3</sup> | (Vacated) TWA: 1 mg/m <sup>3</sup> | REL = 1 mg/m <sup>3</sup> (TWA) |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

OSHA - Occupational Safety and Health Administration

NIOSH: NIOSH - National Institute for Occupational Safety and Health

**Engineering Measures**

Use only under a chemical fume hood. Ensure that eyewash stations and safety showers are close to the workstation location.

Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**  
**Hand Protection**

Goggles  
Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time                 | Glove thickness | Glove comments         |
|----------------|-----------------------------------|-----------------|------------------------|
| Butyl rubber   | See manufacturers recommendations | -               | Splash protection only |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

### **Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced.

To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

No information available.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice.

## 9. Physical and chemical properties

|                                        |                                         |
|----------------------------------------|-----------------------------------------|
| Physical State                         | Solid                                   |
| Appearance                             | Light green                             |
| Odor                                   | Odorless                                |
| Odor Threshold                         | No information available                |
| pH                                     | 2.5 100g/L aq sol (20°C)                |
| Melting Point/Range                    | 105 - 110 °C / 221 - 230 °F             |
| Boiling Point/Range                    | No information available                |
| Flash Point                            | No information available                |
| Evaporation Rate                       | Not applicable                          |
| Flammability (solid,gas)               | No information available                |
| Flammability or explosive limits       |                                         |
| Upper                                  | No data available                       |
| Lower                                  | No data available                       |
| Vapor Pressure                         | No information available                |
| Vapor Density                          | Not applicable                          |
| Specific Gravity                       | No information available                |
| Solubility                             | Soluble in water Alcohol                |
| Partition coefficient; n-octanol/water | No data available                       |
| Autoignition Temperature               | No information available                |
| Decomposition Temperature              | >150 °C                                 |
| Viscosity                              | Not applicable                          |
| Molecular Formula                      | Cl <sub>2</sub> Fe . 4 H <sub>2</sub> O |
| Molecular Weight                       | 198.81                                  |

## 10. Stability and reactivity

|                                         |                                            |
|-----------------------------------------|--------------------------------------------|
| <b>Reactive Hazard</b>                  | None known, based on information available |
| <b>Stability</b>                        | Stable under normal conditions.            |
| <b>Conditions to Avoid</b>              | Incompatible products. Excess heat.        |
| <b>Incompatible Materials</b>           | Strong oxidizing agents                    |
| <b>Hazardous Decomposition Products</b> | Iron oxides, Hydrogen chloride             |
| <b>Hazardous Polymerization</b>         | Hazardous polymerization does not occur.   |
| <b>Hazardous Reactions</b>              | None under normal processing.              |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Category 4.

#### Component Information

| Component                                        | LD50 Oral                | LD50 Dermal               | LC50 Inhalation |
|--------------------------------------------------|--------------------------|---------------------------|-----------------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | ATE LD50 = 705 mg/kg     | Not listed                | Not listed      |
| Iron chloride (FeCl <sub>2</sub> )               | LD50 = 450 mg/kg ( Rat ) | LD50 > 2000 mg/kg ( Rat ) | Not listed      |

**Toxicologically Synergistic Products** No information available

### Delayed and immediate effects as well as chronic effects from short and long-term exposure

|                        |                                                                                          |
|------------------------|------------------------------------------------------------------------------------------|
| <b>Irritation</b>      | Causes burns by all exposure routes                                                      |
| <b>Sensitization</b>   | No information available                                                                 |
| <b>Carcinogenicity</b> | The table below indicates whether each agency has listed any ingredient as a carcinogen. |

| Component                                        | CAS-No     | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|--------------------------------------------------|------------|------------|------------|------------|------------|------------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | 13478-10-9 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Iron chloride (FeCl <sub>2</sub> )               | 7758-94-3  | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** None known

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Product is a corrosive material. Use of gastric lavage or emesis is contraindicated. Possible perforation of stomach or esophagus should be investigated: Ingestion causes severe swelling, severe damage to the delicate tissue and danger of perforation

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

May cause long-term adverse effects in the environment. Do not allow material to contaminate ground water system.

**Persistence and Degradability** based on information available. May persist

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Will likely be mobile in the environment due to its water solubility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

**UN-No** UN3260  
**Proper Shipping Name** CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S.  
**Technical Name** Iron (II) chloride, tetrahydrate  
**Hazard Class** 8  
**Packing Group** III

### TDG

**UN-No** UN3260  
**Proper Shipping Name** CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S.  
**Hazard Class** 8  
**Packing Group** III

### IATA

**UN-No** UN3260  
**Proper Shipping Name** CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S.  
**Hazard Class** 8  
**Packing Group** III

### IMDG/IMO

**UN-No** UN3260  
**Proper Shipping Name** CORROSIVE SOLID, ACIDIC, INORGANIC, N.O.S.  
**Hazard Class** 8  
**Packing Group** III

## 15. Regulatory information

All of the components in the product are on the following Inventory lists: China X = listed Australia U.S.A. (TSCA) Canada (DSL/NDSL) Europe (EINECS/ELINCS/NLP) Australia (AICS) Korea (KECL) China (IECSC) Japan (ENCS) Philippines (PICCS) Taiwan (TCSI) Japan (ISHL) New Zealand (NZIoC) Japan (ISHL)

### International Inventories

| Component                                        | CAS-No     | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|--------------------------------------------------|------------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | 13478-10-9 | -   | -    | -    | -                                             | -         | -      | -   |
| Iron chloride (FeCl <sub>2</sub> )               | 7758-94-3  | X   | -    | X    | ACTIVE                                        | 231-843-4 | -      | -   |

| Component                                        | CAS-No     | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|--------------------------------------------------|------------|-------|----------|------|------|------|------|-------|-------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | 13478-10-9 | X     | -        | X    | X    | X    | X    | X     | X     |
| Iron chloride (FeCl <sub>2</sub> )               | 7758-94-3  | X     | KE-21085 | X    | X    | X    | X    | X     | X     |

### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

## Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

## Other International Regulations

Authorisation/Restrictions according to EU REACH

Not applicable

## Safety, health and environmental regulations/legislation specific for the substance or mixture

| Component                                        | CAS-No     | OECD HPV | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|--------------------------------------------------|------------|----------|------------------------------|---------------------------|--------------------------------------------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | 13478-10-9 | Listed   | Not applicable               | Not applicable            | Not applicable                             |
| Iron chloride (FeCl <sub>2</sub> )               | 7758-94-3  | Listed   | Not applicable               | Not applicable            | Not applicable                             |

| Component                                        | CAS-No     | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|--------------------------------------------------|------------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| Iron chloride (FeCl <sub>2</sub> ), tetrahydrate | 13478-10-9 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Iron chloride (FeCl <sub>2</sub> )               | 7758-94-3  | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

### Prepared By

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### Creation Date

04-February-2010

### Revision Date

20-September-2024

### Print Date

20-September-2024

### Revision Summary

This document has been updated to comply with the requirements of WHMIS 2015 to align with the Globally Harmonised System (GHS) for the Classification and Labelling of Chemicals.

## Disclaimer

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**